BMJ Health & Care Informatics

Data-driven queueing modelling: a simulation case study of emergency department crowding

Adrien Wartelle ,[1] Farah Mourad-Chehade,[2] Farouk Yalaoui,[2] David Laplanche,[3] Stephane Sanchez[3,4]

To cite: Wartelle A, Mourad-Chehade F, Yalaoui F, *et al*. Data-driven queueing modelling: a simulation case study of emergency department crowding. *BMJ Health Care Inform* 2026;**33**:e101462. doi:10.1136/bmjhci-2025-101462

► Additional supplemental material is published online only. To view, please visit the journal online (https://doi.org/10.1136/bmjhci-2025-101462).

Received 08 February 2025
Accepted 27 December 2025

Check for updates

© Author(s) (or their employer(s)) 2026. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

[1]IRIT, Université de Toulouse, CNRS, Toulouse INP, INU Champollion, Toulouse, France
[2]Computer Science and Digital Society Laboratory (LIST3N), Université de Technologie de Troyes, Troyes, France
[3]Public Health and Performance Department, Centre Hospitalier de Troyes, Troyes, France
[4]Research on Health, URCA, Reims, France

Correspondence to
Dr Adrien Wartelle;
adrien.gj.wartelle@gmail.com

ABSTRACT

Objectives Emergency department crowding refers to a complex state of congestion associated with a set of performance indicators such as occupation levels, waiting times and specific scores. Among current methods to model it, an objective gap exists between forecasting machine learning methods, focusing on prediction precision and queueing and simulation methods, focusing on capturing correctly the effect of decision variables for evaluation and optimisation purposes. The objective of the present analysis is to implement and numerically validate a novel data-driven queueing methodology that can bridge this gap and to show its applicability in a simulation case study.

Methods A statistical modelling of the queueing processes, particularly patient departure rates and probabilities, is developed to cross the gap defined above. Using the data from a major emergency department of eastern France, the resultant data-driven queueing network model is validated and applied through a synchronous simulation algorithm.

Results The model obtained considers the complex effects of patient arrivals and doctor and nurse allocations while offering an unbiased and accurate measure of long-term crowding. Its application with the case study quantifies the impact of the opening of new Unscheduled Care Services on emergency department crowding.

Discussion The new data-driven queueing methodology is able to model and quantify complex crowding effects at a detailed level in an emergency department.

Conclusions This study shows an alternative approach successfully bridging the modelling gap by establishing a model that can effectively predict system crowding dynamics under the influence of multiple key variables.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Studies on the impact of Unscheduled Care Services on the emergency department have been investigated in terms of arrival, patient flow, entry numbers and costs, but their effects on emergency department crowding have not been analysed.

WHAT THIS STUDY ADDS

⇒ A novel data-driven queueing modelling methodology is implemented to quantify the impact of Unscheduled Care Services on emergency department crowding. This powerful model crosses a gap between statistical and simulation methods by performing this evaluation within a data-driven framework with real world data.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This study furnishes healthcare regulators with an alternative and feasible approach, compared with standard controlled ones, to quantify the impacts of the new health services structures and have evaluation starting point regarding their ability to redirect patient flows and optimise congestion. Furthermore, the new data-driven queueing offers many possible applications to other emergency departments and systems.

INTRODUCTION

The modelling of service and production systems is a central topic in operational research. To provide adequate decision support, the models produced must be able to capture correctly the behaviour of a system. This adequation is judged in their capacity to reproduce the variations of selected key performance indicators (KPIs). In the healthcare system context, the key notion of crowding, used to characterise the state of an emergency department (ED), encapsulates most of the KPIs targeted by these models. It refers to a complex state of congestion of a service system which is not limited to occupation level and can also be measured through a set of other KPIs such as the waiting time, the proportion of left without being seen and even more tailored crowding scores.[1–3] Among the solutions to crowding, Unscheduled Care Services (UCS) encompass a whole range of healthcare structures for patients requiring immediate care, notably offering alternative pathways from the ED for low-acuity patients.[4] Many studies have been performed to evaluate the impact of their opening near a targeted ED, as shown in online supplemental material A

BMJ Health & Care Informatics: first published as 10.1136/bmjhci-2025-101462 on 8 January 2026. Downloaded from https://informatics.bmj.com on 15 January 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

and in a previous study,[5] notably on the volume of visits avoided and their associated costs but not on the associated crowding avoided.[6] To perform this type of evaluation, an ED patient flow model is required.[7–9] The review of Bhattacharjee and Ray describes the different components of this type of model with the different aspects to be modelled, for example, the network structure, the patient arrival distribution, the branching or transition probabilities and the service time distribution.[7] An ED patient flow model decomposes also the different modelling approach into analytic models, with queueing models and Markov chains and compartmental models, simulation models, with discrete-event and agent-based models,[9] and statistical/empirical models. In the context of the ED, reviewed by Badr *et al*, Hu *et al* and Ouda *et al*,[1 8 9] these different approaches are used for different levels of crowding management objectives.

First, statistical and empirical models are used for diagnosing crowding using scores, data and process-mining of patient electronic health records, and for forecasting crowding.[10–15] Second, analytical and simulation models are used for evaluating and then optimising crowding dynamics conjointly with other variables such as the volume of patient arrivals, the dimensioning and scheduling of resources, and the occurrence of external events.[16–26] An objective gap can be observed between these two different types of crowding-model approach, the former focusing on the prediction accuracy of the model using a data-driven approach, the latter focusing on integrating and capturing correctly the effects of decision and environmental variables on the prediction at the cost of lower accuracy. Consequently, the objective of this study is to implement and validate numerically a novel data-driven queueing simulation model that can bridge this gap and show its applicability through a simulation case study regarding the impact of alternative UCSs on ED crowding.

MATERIAL AND METHODS

The model approach presented here is inspired by the work of Whitt and Zhang which presented a data-driven model of the ED for forecasting patient arrivals and occupation level based on arrival rate and probability of departure models.[12] To extend this approach, a simulation model of a data-driven queueing network is presented here based on the statistical modelling of departure rates of patients.

Modelling problem formulation

The crowding of a service system, captured by different measures, is governed by its arrival process $A(t)$ and its departure process $D(t)$ quantifying the number of patient arrivals and departures over time. Considering the existing literature,[1 3 27] different measures were selected, including the occupancy level $l(t) = A(t) - D(t)$, $t \in [t_1, t_2]$, the average occupancy level $L_h(t) = \int_{t-h}^{t} A(u) - D(u)\, du$, the approximate waiting time at departure $\widetilde{W}_{d,h} = \frac{1}{D(t)-D(t-h)} \int_{t_1}^{t_2} A(u) - D(u)\, du$ (related to Little's law $L = \lambda\, W$)[28] and the recently introduced congestion $\rho_h(t) = \frac{\int_{t-h}^{t} A(u)-D(t-h)\, du}{\int_{t-h}^{t} D(u)-D(t-h)\, dt}$ validated in a previous study.[3] To account for the complex internal behaviour of the system when modelling the behaviour of these two processes, this study uses a patient flow model breaking down the ED system into relevant stations associated to a decomposition in time. The main notions used in the model are listed in table 1. Further details on the model and its simulation algorithm are also available in online supplemental material B.

The global arrival process $A(t)$ and departure process $D(t)$ evolution, and thus also the crowding measures evolution, are modelled through $A_{0,n}$ and $D_{o,n}$ which can be deduced from all others $A_{i,n}$ and $D_{i,n}$, with $n = \left\lceil \frac{t}{1\ min} \right\rceil$ the discretised indexed time.

Modelling solution proposed

The model approach proposed in this study relies on the modelling of initial arrivals $a_{0,n}$, circuits of patients u_m and departures $d_{i,n}$. These rate and probability machine-learning models are all trained on real world data and then used in a synchronous simulation algorithm described in online supplemental material B and illustrated in figure 1 to get an estimation of $A(t)$ and $D(t)$ and thus of ED crowding.

Initial arrivals are modelled for each class of patient c_m with $a_{0,n} = \sum_{c_m \in \{0,1\}} a_{0,c_m,n}$ and using non-homogeneous Poisson processes, with Poisson distribution $\mathcal{P}$, whose intensity $\Lambda_{0,c_m}(n)$ is a function of time $n \in N$:

$$a_{0,c_m,n} \sim \mathcal{P}\left(\Lambda_{0,c_m}\left(\mathrm{HD_n}, \mathrm{DW_n}, \mathrm{Z_n}\right)\right), \forall n \in N \quad (1)$$

Patients' circuits u_m are modelled using logistic regression to determine the probabilities of certain categories of circuit based on the arrival time ta_m, and using historical averages to determine the exact probabilities of each potential circuit.

At the heart of the original data-driven model, the station departures $d_{i,n}$ are generated sequentially through departures $d_{i,n|(n-1)}$ with the knowledge of system state at time $(n-1)$. Those are modelled through the patient instantaneous departure posterior probabilities $p_{i,n,m|(n-1)} = P\left(td_{i,m} = n | td_{i,m} \geq (n-1)\right)$ using a Binomial $\mathcal{B}$ and Poisson Binomial $\mathcal{PB}$ distributions:

$$d_{i,n,m|(n-1)} \sim \mathcal{B}\left(p_{i,n,m|(n-1)}\right) \quad (3)$$

$$d_{i,n|(n-1)} \sim \mathcal{PB}\left(\left(p_{i,n,m|(n-1)}\right)_{m \in N}\right) \quad (4)$$

Each variable $p_{i,n,m|(n-1)}$ is modelled as a rate of departure μ_i which is a function of time, patient state and system external crowding factors such as previous arrivals and the number of actors:

$$p_{i,n,m|(n-1)} = \mu_i\left(\left(A_{i,n'}\right)_{n'<n}, la_{i,n}, c_m, w_{i,n,m}, HD_n, DW_n, Z_n\right) \quad (5)$$

To estimate each function μ_i, a binomial counting table of all $d_{i,n,m}$ values is extracted for discredited instant n

BMJ Health & Care Informatics: first published as 10.1136/bmjhci-2025-101462 on 8 January 2026. Downloaded from https://informatics.bmj.com on 15 January 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

Table 1 Mathematical framework description

Mathematical notion	Description
$\mathbf{n} \in \mathbf{N}$	Indexed time (1 min scale)
$(HD_n,\ DW_n, Z_n)\ ,\ \forall n \in \mathbb{N}$	Temporal indicators of the hour of the day, the day of the week and 6-hour zone of the day
$m \in \mathbb{N}$	Patient indices given in order of arrival in the ED
$c_m \in \{0,1\}$	Class indicator of a patient (ie, 1 if they belong to a cluster affected by UCS[5], else 0) during their entire stay
$i \in I$	Indices of stations which characterise the ED decomposition in 5 phases of care which are triage, initial examination in short circuit (SC) and in long circuit (LC), supplementary examinations in SC and in LC (and 0 for the global system)
$ta_{i,m}, td_{i,m} \in \mathbb{R}^+$	Time of arrival and departure of patient *m* in station *i*
$w_{i,n,m} \equiv n - ta_{i,m},$ $ta_{i,m} \leq n \leq td_{i,m}$	Current waiting time of patient *m* at time *n* in station *i*
$u_{m,k,}k \in [1, K_m]$ $u_m = \{u_{m,k}\}_{(1 \leq k \leq K_m)}$	Circuit of patient *m*, ie, a list of unique station indexes of size K_m describing the set of stations that a patient will visit in order
$la_{i,n}$	Current number of actors (doctors or nurses)
$A_{i,n} \equiv \sum_{\mathbf{n}' \leq \mathbf{n}} a_{i,n'}$ $a_{i,n} \equiv A_{i,n} - A_{i,n-1} \equiv \sum_{m \in N} a_{i,n,m}$ $a_{i,n,m} \equiv \mathbb{1}_{ta_{i,m}=n}$ $D_{i,n} \equiv \sum_{n' \leq \mathbf{n}} d_{i,n'}$ $d_{i,n} \equiv D_{i,n} - D_{i,n-1} \equiv \sum_{m \in N} d_{i,n,m}$ $d_{i,n,m} \equiv \mathbb{1}_{td_{i,m}=n}$	Stochastic arrival $A_{i,n}$ and departure $D_{i,n}$ processes of stations and their decomposition per time step *n* and per patient *m*
$l_{i,n} = A_{i,n} - D_{i,n}$	Occupation level of station *i* at time *n*

ED, emergency department; UCS, Unscheduled Care Services.

of each patient m during its presence in one station i of the system along with the set of independent variables described in (5). The model selection process, described and performed in online supplemental material B and C is made between several methods and formulation using generalised Poisson linear regression, Poisson tree-based regression and Poisson random forest.

Study design and context

The study evaluation is based on the extraction of 34 358 ED visit records, representing 99.96% of all visits in the study period, and the associated time-registered locations and prescriptions occurring during the visits, in the period from 11 October 2018 to 1 May 2019 of Troyes ED, a major ED in Eastern France with an important arrival volume of more than 60 000 patients per year.[29] As detailed in online supplemental material B, these data are processed to obtain the observed values of each independent variable $\left(A_{i,n'}\right)_{n' \leq n}$, $la_{i,n}$, c_m, $W_{i,n,m}$, HD_n, DW_n and Z_n and each dependent variable $d_{i,n,m|(n-1)}$, a_0, c_m, n, u_m by mapping the data of each visit to the stations i and the time indexes n, for each index m of the 34 358 patient visits.

Using this processed data, this study evaluation is composed of two parts. The first part regards the evaluation of the probability departure models μ_i proposed for each station i with their ability to reproduce the crowding measure and effects observed in data. The second part is the simulation case study on the potential impact the launching of two new UCS, opened during the first week of October 2018, on the crowding of the ED of Troyes. This part is a direct extension of a previous study which focused on the detailed impact on arrival flow of patients through multimorbidity clustering of patient visit profiles and their diagnoses.[5 30] This impact is translated in terms of the modification of the arrival pattern of class 1 patients, ie, patients belonging to a cluster affected by UCS,[5 30] and in turn in terms of crowding avoided for the ED. This evaluation design is detailed further in online supplemental material B.

RESULTS

During the 28-week study period from 14 October 2018 to 28 April 2019, 34 358 patients arrived in the ED of Troyes, with 5627 from class 1 (16.4%) and 28 731 from class 0

BMJ Health & Care Informatics: first published as 10.1136/bmjhci-2025-101462 on 8 January 2026. Downloaded from https://informatics.bmj.com on 15 January 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

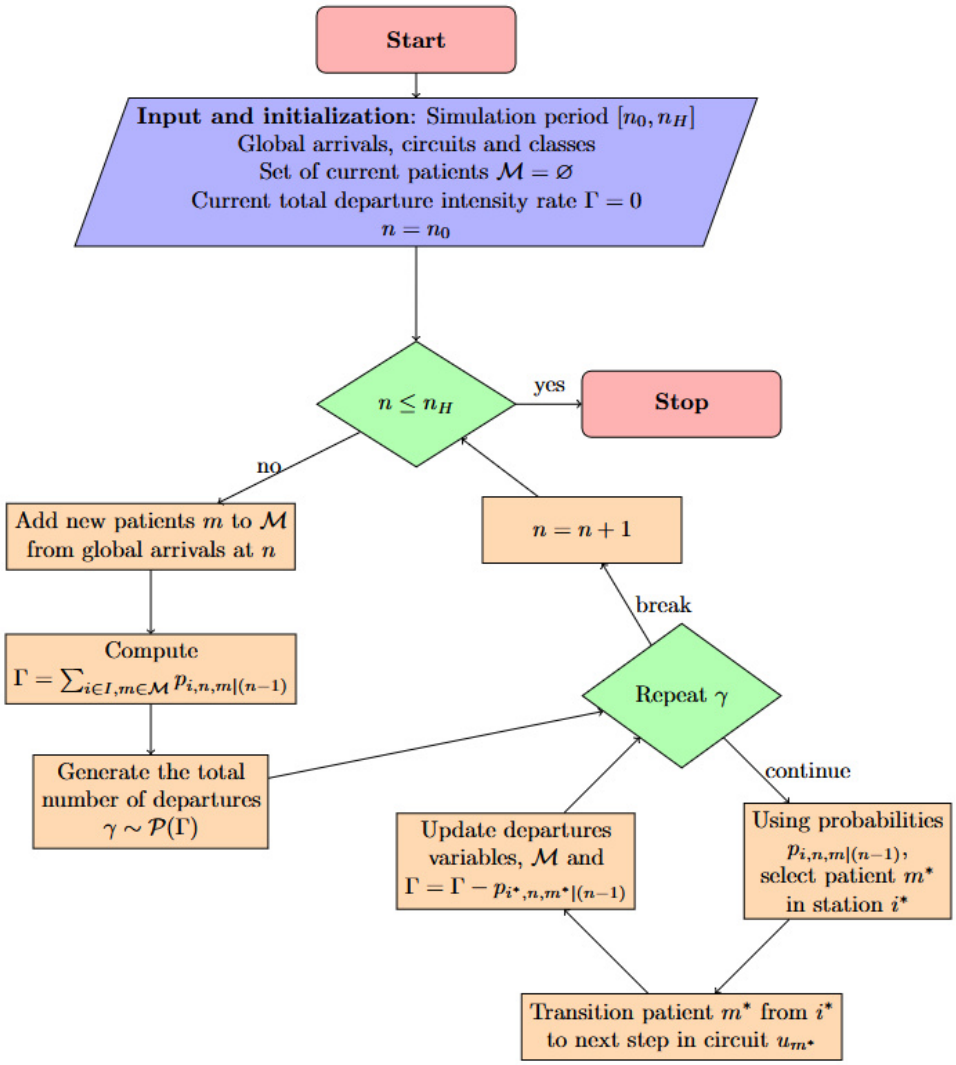


Figure 1 Synchronous simulation algorithm (diagram). The asterisk in m* and i* are used to specify that a particular value of m and i have been selected whereas the indices used in probabilities p_{i,n,m}|(n-1) refer to any m, n and i.

(83.6%). From these, 554 (9.8%) of class 1 patients went in the long circuit compared with 10 596 (36.9%) for class 0 patients. The global average waiting times observed are 4 hours 36 min (resp. 4 hours 46 min for class 0 and 3 hours 41 min for class 1) and the global average occupation 32.5. The average waiting time in triage was 25.6 min, in long circuit 7 hours 12 min and in short circuit 2 hours 44 min.

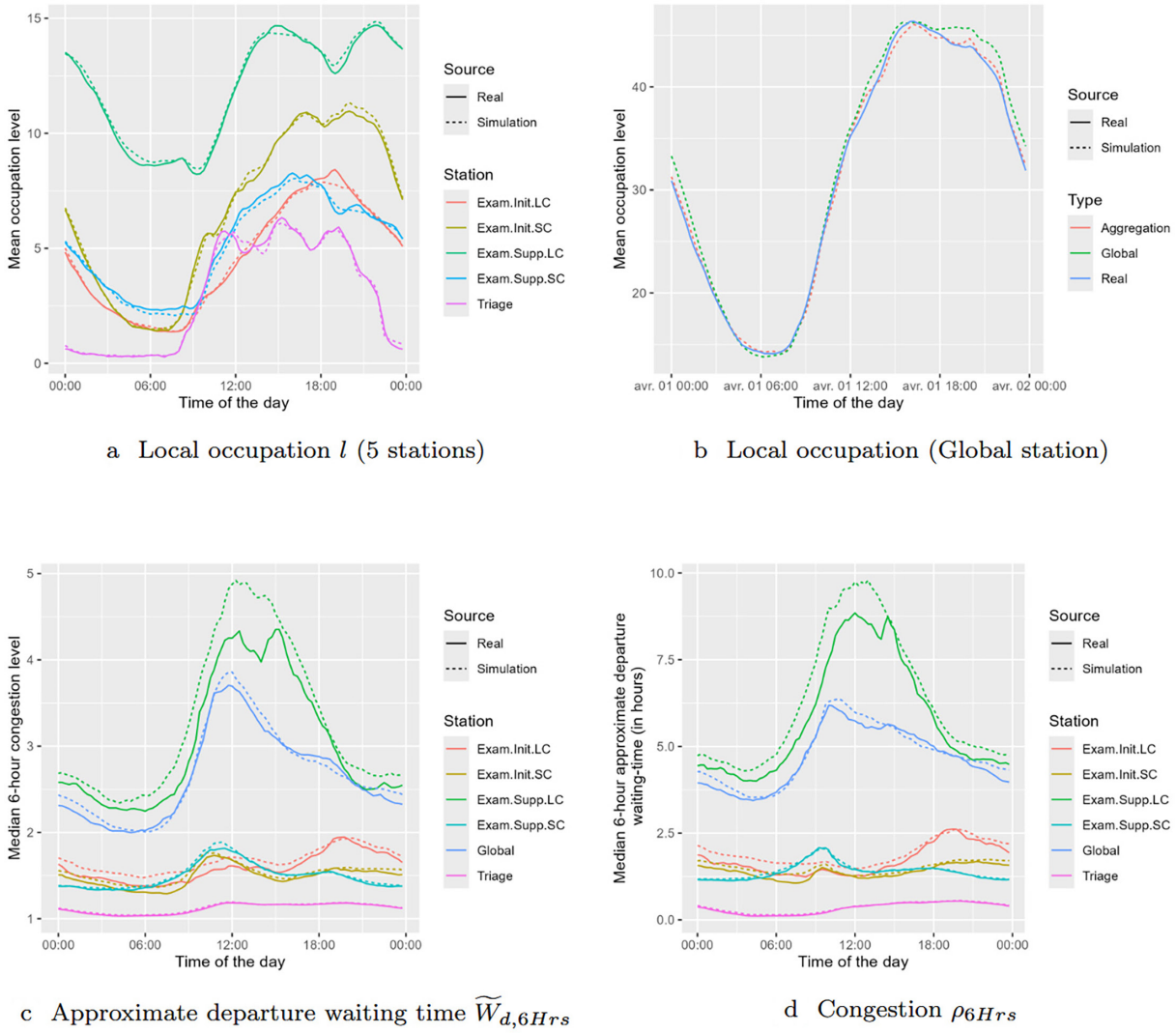
Evaluation of the probability departure models

A Poisson regression departure model was selected through the quality analysis, available in online supplemental material C, of the different estimation methods and their goodness of fit with departure data knowing patient arrivals. Specifically, with arrivals and circuit data known, the selected departure model (id 9) can estimate, on average for each station, the local occupation level with a mean absolute relative error (MARE) of 26.0% and the 6-hour mean occupation level with a MARE of 16.3%, improving from the 39.1% and 29.7% from the constant departure rate model (id 1). Figure 2 illustrates the mean values of different crowding measures, aggregated over the 28 weeks, during the different times of the day with the selected model simulation and with real observations. As suggested by figure 2b, the total bias error of the model on global occupation l is very low with a value of 0.33 (1.01%) for the estimation using a one-station modelling approach and 0.04 (0.14%) using the aggregation of the five-station approach. Over the study period, the *MARE* value on the mean aggregate local occupation throughout the day, averaged on the 28 weeks, is 2.4% for the one-station approach and 1.3% for the five-station approach. As illustrated by figure 2a, the MARE values on local occupation are all below 3.6% for each individual station. For the other crowding measures, illustrated in figure 2c,d using median values, the *MARE* on mean aggregate of the 5 stations is 7.56% for approximate departure waiting time with a lag of 6 hours $\widetilde{W}_{d,6Hrs}$ and 3.67% for the congestion measure with the same lag ρ_{6Hrs}.

Application case study: impact of the UCS on ED crowding

As mentioned in the modelling solution section, the simulation case study relies on the modelling of arrivals

BMJ Health & Care Informatics: first published as 10.1136/bmjhci-2025-101462 on 8 January 2026. Downloaded from https://informatics.bmj.com on 15 January 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.



a Local occupation l (5 stations)

b Local occupation (Global station)

c Approximate departure waiting time $\widetilde{W}_{d,6Hrs}$

d Congestion ρ_{6Hrs}

Figure 2 Mean value throughout the day of different crowding measures. Except for the red curve for global station 0 in (b), all stations are modelled individually. LC, long circuit; SC, short circuit.

and circuits of patients of class 0 and 1 which are detailed in online supplemental material C. The different hypotheses of the UCS impact hypothesis in terms of arrivals per week of class 1 patients are illustrated in figure 3a with different values of relative volume increase α_{arrl}:

- $\alpha_{arrl} = 1$ considers that no decrease of arrivals occurred due to the UCS opening.
- $\alpha_{arrl} = 1.21$ considers the volume of arrivals would be equal to the average of the 3 last years (2015–2017) on the same months period as the study.
- $\alpha_{arrl} \in \{1.23, 1.27, 1.32\}$ considers, respectively, the minimum, mean and maximal linear extrapolation of the 3 last years' volume.

The other figures show the effect on relative occupation and waiting time of the impact, for all stations in the five-station approach, with the scenario of α_{arrl}=1.27 taken as a reference. Globally, this results in an occupation increase of 5% for the system considering the full five-station queueing network approach, used in figure 3c,d, and 4% for the approach considering one global station. Using Little's law relation and considering a fixed waiting time W, one can compute $\frac{L_{\alpha_{arrl}=1.27}}{L_{\alpha_{arrl}=1}} = \mathbb{P}(c_m = 0) + \mathbb{P}(c_m = 0) \times 1.27 = 0.836 + 0.164 \times 1.27 = 1.0443$, that is, a 4.44% increase. Accounting for the 0.627% global increase for W indicated in table 2, one has $\frac{L_{\alpha_{arrl}=1.27}}{L_{\alpha_{arrl}=1}} = (\mathbb{P}(c_m = 0) + \mathbb{P}(c_m = 0) \times 1.27) \times 1.00627 = 1.050848$, that is, a *5.08%* increase more coherent with the five-station approach. This reasoning can be applied for all stations, particularly for the Triage station where the impact in occupation level is more due to the indirect effect of the 4.64% increase in W than the 4.43% increase in arrival rate.

A simulation experiment, detailed in online supplemental material C, was performed to translate the impact in terms of number of actors avoided. Applying a factor $\alpha_{act} \in (0.5, 2)$ and using the five-station approach, a local linear model showed that the impact of the UCS with α_{arrl}=1.27 in terms of waiting time would correspond to a 33.1% increase (α_{act}=1.331) avoided in the mean number

BMJ Health & Care Informatics: first published as 10.1136/bmjhci-2025-101462 on 8 January 2026. Downloaded from https://informatics.bmj.com on 15 January 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

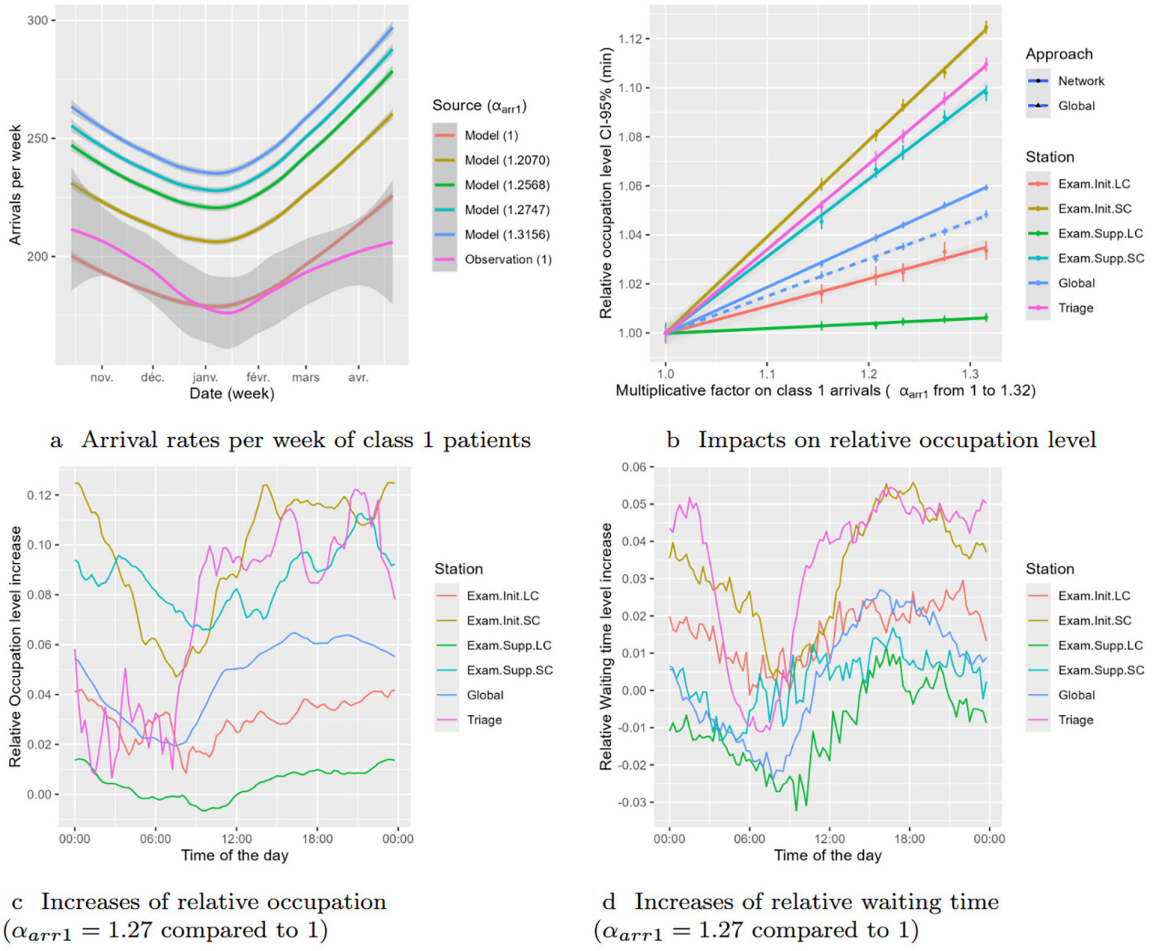


Figure 3 Illustration of the impacts of UCS based on α_arr1 hypotheses. LC, long circuit; SC, short circuit; UCS, Unscheduled Care Services.

of nurses in Triage and to a 12.8% increase (α_{act}=1.128) avoided in the mean number of doctors in Exam.init.SC.

DISCUSSION

The queueing modelling approach proposed by this study is the culmination of an advanced strategy of crowding evaluation proposing a comprehensive data-driven characterisation of a complex system such as the ED and its patient flow.[3 5 30] It completes the step-by-step approach begun in previous patient flow studies through a machine-learning characterisation of the service performances and its evaluation with a synchronous simulation algorithm. According to our knowledge, and discussed further in the online supplemental material D, it is the first study to perform this characterisation in such a manner, associated with an in-depth statistical evaluation of model quality regarding its ability to capture crowding dynamics. Due to the requirement to rely on

Table 2 Waiting time impact of UCS with five-station approach (Average waiting time with 95% CI on 10 replications per parameter)

Station	*W*±1.96×*SD* (α_{arr1} = 1)	W'±1.96×SD (α_{arr1} = 1.27)	Difference (%)
Global	4*h*14*m*54*s*±14*s*	4*h*16*m*30*s*±14*s*	+1*m*36*s* (+0.627%)
Triage	0*h*26*m*34*s*±3*s*	0*h*27*m*48*s*±3*s*	+1*m*14*s* (+4.64%)
Exam.Init.LC	2*h*01*m*54*s*±22*s*	2*h*03*m*57*s*±19*s*	+2*m*03*s* (+1.68%)
Exam.Supp.LC	5*h*22*m*57*s*±35*s*	5*h*19*m*28*s*±35*s*	–3*m*29*s* (+1.07%)
Exam.Init.SC	1*h*29*m*52*s*±7*s*	1*h*33*m*12*s*±8*s*	+3*m*20*s* (+3.71%)
Exam.Supp.SC	1*h*19*m*40*s*±14*s*	1*h*19*m*48*s*±10*s*	+8*s* (+0.167%)

h, hours; LC, long circuit; m, min; s, second; SC, short circuit; UCS, Unscheduled Care Services.

BMJ Health & Care Informatics: first published as 10.1136/bmjhci-2025-101462 on 8 January 2026. Downloaded from https://informatics.bmj.com on 15 January 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

systematically recorded available data, this model has difficulties in capturing as many real-world nuances as agent-based and discrete-event simulation models, such as the assignment of patients to nurses and doctors and the detail sequences of actions performed by them to provide care.[9 10] Nevertheless, with sufficient improvement of recording coverages and quality, the data-driven model can approach these levels of nuances through its flexible machine learning modelling formulations while proposing a statistically unbiased and accurate estimation of crowding measures suitable for forecasting, evaluation and optimisation purposes. These formulations rely on the estimation of the key variable of patient instantaneous departure rate μ_i through the probabilities $p_{i,n,m|(n-1)}$, with respect to general variables on clients, resources and time that are the main contributors to crowding behaviours and that make possible the extension of the model to any other ED and any type of queueing system without any modification, requiring only the availability of temporal records of these variables.

The study is part of a wider one on the impact of opening UCS within a healthcare system on patient flow, particularly that of ED.[5 30] Although this is not the first behavioural study on congestion,[17] this simulation study is the first to address the impact of UCS by identifying the effect of the changes in arrival patterns on ED crowding while considering the complex ED service behaviours. The effect found here in terms of crowding was moderate, with mainly 5% occupation level increase avoided globally, 4.64% of waiting time in Triage and 3.71% in Exam.Supp. SC, but representing a more important 33.1% increase in the mean number of avoided Triage nurse and 12.8% doctor in Short Circuit doctors. These observations can be explained primarily by the fact that the impacted group is almost exclusively (over 90%) composed of short-circuit visits and that the change in average occupancy is the compounded result of changes in arrival numbers and waiting times, as shown by Little's law $L = \lambda W$.[24] The initial result on partial numbers of actors avoided gives a good scale of the effect but must be verified with a transposition to realistic configurations of whole quantities while respecting scheduling constraints on the placement and duration of shifts for doctors and nurses.[25]

CONCLUSIONS

Through the new data-driven modelling approach, the crowding evaluation performed in this study allows healthcare regulators to understand the impact of UCS on ED crowding and can be used as a starting evaluation point, alongside a deeper focus on health and economic aspects, to make recommendations on the future implementation of UCS. The standard approach using controlled and randomised experiments for this type of opening effect is nearly impractical due to the costs involved and the national scale required for intervention. Thus, in this context, the approach presented here proposes a retrospective observational study targeting a specific area that, despite its limitations, represents a crucial step in evaluating UCS as a solution to ED crowding.

Future work should expand the modelling approach on other systems, in healthcare and other industries, and for other purposes such as development of advanced predictive and forecasting applications for healthcare regulators to make informed decisions, as is currently the case with the development of a simulation software based on the study's model for the ED of Troyes. The expansion to other systems, requiring a change in the design of stations and patients' classes, will evaluate the flexibility and robustness of the approach, validate it further and reduce its shortcomings through several avenues. On the model itself, the first would be to incorporate more detailed variables on the system while limiting potential approximation biases. It would be possible to combine the data-driven approach with a more traditional simulation modelling approach when data are unavailable, for example, when exploring a new system design that does not yet exist. Second, the distribution hypothesis on the different processes of arrivals, departures and circuits should be challenged to consider potential excess of variability that could not be captured with a Poisson distribution, its independence hypothesis and its resulting coefficient of variation of 1.

Software

The data used in this study were extracted using the Troyes ED database software, named RESURGENCES and using SQL requests through the SQL developer software. This software is directly used by ED physicians and residents to register patients' data during their stay. The data were analysed using R V.4.0.2 in the RStudio V.1.2.5042 environment with the packages tidyverse V.1.3.0, lubridate V.1.7.9 and stats V.4.0.2. All data transformations as well as simulation (see online supplemental material B) were performed using the C++ language, which is now used for a software prototype for the Troyes Hospital Center.

Contributors AW is responsible for the overall content as guarantor. AW conceived the simulation model formulation, implemented it, applied it to the simulation case study and wrote the paper. FM-C, FY, DL and SS were the scientific advisors supervising the course of the study. Specifically, FM-C and FY contributed to the critic of simulation modelling approach, and DL and SS to the simulation case study while enabling the data collection from the Hospital of Troyes. SS also contributed to reviewing the revisions made on the manuscript.

Funding The authors have not declared a specific grant for this research from any funding agency in the public, commercial or not-for-profit sectors.

Competing interests None declared.

Patient consent for publication Not applicable.

Ethics approval This study was performed in compliance with the national legislation regarding epidemiological studies (Declaration No 2203674v0 of 24 July 2018). Since the study was wholly observational and only used anonymised data (patient names were not recorded), neither ethics approval nor specific written informed consent from participants was required under French law for a retrospective database study. Moreover, in accordance with national ethical directives, the requirement for written informed consent was waived because the study was strictly observational, and all data were blinded. (Ref: French Public Health Code. Article R. 1121-2.) According to the French Public Health Code, this research also did not require an ethics committee (Ref.: French Public Health Code. Article R. 1121-2.). The study was conducted with all medical information-related

BMJ Health & Care Informatics: first published as 10.1136/bmjhci-2025-101462 on 8 January 2026. Downloaded from https://informatics.bmj.com on 15 January 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.

legislation and was declared to the National Registry of Health research body under No. 1113130319. Patients were informed that the study was being carried out via the hospital's registry of ongoing studies.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Due to their confidential nature protected by French law (see Ethical statement), the health data exploited here, taken from the Troyes Hospital RESURGENCES software from 2017 to 2022, cannot be made available.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: https://creativecommons.org/licenses/by-nc/4.0/.

ORCID iD
Adrien Wartelle https://orcid.org/0000-0003-1471-0866

REFERENCES

1 Badr S, Nyce A, Awan T, *et al*. Measures of Emergency Department Crowding, a Systematic Review. How to Make Sense of a Long List. *Open Access Emerg Med* 2022;14:5–14.
2 Eyre A, Stein GY, Chen J, *et al*. Physician performance scores used to predict emergency department admission numbers and excessive admissions burden. *BMJ Health Care Inform* 2024;31:e101080.
3 Wartelle A, Mourad-Chehade F, Yalaoui F, *et al*. Changing the perspective of system crowding evaluation using a new congestion measure: application to the Emergency Department. *Oper Res Int J* 2024;24:53.
4 Morley C, Unwin M, Peterson GM, *et al*. Emergency department crowding: A systematic review of causes, consequences and solutions. *PLoS One* 2018;13:e0203316.
5 Wartelle A, Mourad-Chehade F, Yalaoui F, *et al*. Multimorbidity clustering of the emergency department patient flow: Impact analysis of new unscheduled care clinics. *PLoS One* 2022;17:e0262914.
6 Moe J, Oland R, Moe G. Impact of a Primary Care After-Hours Clinic on Avoidable Emergency Department Visits and Costs. *Healthc Q* 2019;22:42–7.
7 Bhattacharjee P, Ray PK. Patient flow modelling and performance analysis of healthcare delivery processes in hospitals: A review and reflections. *Comput Ind Eng* 2014;78:299–312.
8 Hu X, Barnes S, Golden B. Applying queueing theory to the study of emergency department operations: a survey and a discussion of comparable simulation studies. *Int Trans Operational Res* 2018;25:7–49.
9 Ouda E, Sleptchenko A, Simsekler MCE. Comprehensive review and future research agenda on discrete-event simulation and agent-based simulation of emergency departments. *Simul Model Pract Theory* 2023;129:102823.
10 Cho M, Song M, Park J, *et al*. Process Mining-Supported Emergency Room Process Performance Indicators. *Int J Environ Res Public Health* 2020;17:6290.
11 Whitt W, Zhang X. A data-driven model of an emergency department. *Oper Res Health Care* 2017;12:1–15.
12 Whitt W, Zhang X. Forecasting arrivals and occupancy levels in an emergency department. *Oper Res Health Care* 2019;21:1–18.
13 Cheng Q, Argon NT, Evans CS, *et al*. Forecasting emergency department hourly occupancy using time series analysis. *Am J Emerg Med* 2021;48:177–82.
14 Sharafat AR, Bayati M. PatientFlowNet: A Deep Learning Approach to Patient Flow Prediction in Emergency Departments. *IEEE Access* 2021;9:45552–61.
15 Jiang S, Liu Q, Ding B. A systematic review of the modelling of patient arrivals in emergency departments. *Quant Imaging Med Surg* 2023;13:1957–71.
16 Weiss SJ, Derlet R, Arndahl J, *et al*. Estimating the degree of emergency department overcrowding in academic medical centers: results of the National ED Overcrowding Study (NEDOCS). *Acad Emerg Med* 2004;11:38–50.
17 Ahalt V, Argon NT, Ziya S, *et al*. Comparison of emergency department crowding scores: a discrete-event simulation approach. *Health Care Manag Sci* 2018;21:144–55.
18 Ahmed MA, Alkhamis TM. Simulation optimization for an emergency department healthcare unit in Kuwait. *Eur J Oper Res* 2009;198:936–42.
19 Yousefi M, Yousefi M, Fogliatto FS. Simulation-based optimization methods applied in hospital emergency departments: A systematic review. *Simulation* 2020;96:791–806.
20 Cochran JK, Bharti A. A multi-stage stochastic methodology for whole hospital bed planning under peak loading. *Int J Ind Syst Eng* 2006;1:8.
21 Green LV, Soares J, Giglio JF, *et al*. Using queueing theory to increase the effectiveness of emergency department provider staffing. *Acad Emerg Med* 2006;13:61–8.
22 Green LV, Kolesar PJ, Whitt W. Coping with Time-Varying Demand When Setting Staffing Requirements for a Service System. *Prod Oper Manag* 2007;16:13–39.
23 Cochran JK, Roche KT. A multi-class queuing network analysis methodology for improving hospital emergency department performance. *Comput Oper Res* 2009;36:1497–512.
24 de Bruin AM, Bekker R, van Zanten L, *et al*. Dimensioning hospital wards using the Erlang loss model. *Ann Oper Res* 2010;178:23–43.
25 El-Rifai O, Garaix T, Augusto V, *et al*. A stochastic optimization model for shift scheduling in emergency departments. *Health Care Manag Sci* 2015;18:289–302.
26 Yankovic N, Green LV. Identifying Good Nursing Levels: A Queuing Approach. *Oper Res* 2011;59:942–55.
27 Pearce S, Marchand T, Shannon T, *et al*. Emergency department crowding: an overview of reviews describing measures causes, and harms. *Intern Emerg Med* 2023;18:1137–58.
28 Little JDC, Graves SC. Little's law. In: *International series in operations research and management science*. 2008: 81–100.
29 Naouri D, El Khoury C, Vincent-Cassy C, *et al*. The French Emergency National Survey: A description of emergency departments and patients in France. *PLoS One* 2018;13:e0198474.
30 Wartelle A, Mourad-Chehade F, Yalaoui F, *et al*. Clustering of a Health Dataset Using Diagnosis Co-Occurrences. *Appl Sci (Basel)* 2021;11:2373.

BMJ Health & Care Informatics: first published as 10.1136/bmjhci-2025-101462 on 8 January 2026. Downloaded from https://informatics.bmj.com on 15 January 2026 by guest. Protected by copyright, including for uses related to text and data mining, AI training, and similar technologies.